The Honorable Richard A. Jones

UNITED STATES DISTRICT COURT
WESTERN DISTRICT OF WASHINGTON
AT SEATTLE

ROYAL J. O'BRIEN, Plaintiff, v. MICROSOFT CORPORATION, Defendant.	NO. 2:19-cv-01625-RAJ **DECLARATION OF WILLIAM ANTHONY MASON IN SUPPORT OF PLAINTIFF'S RESPONSE TO MICROSOFT CORPORATION'S MOTION TO DISMISS COMPLAINT UNDER RULE 12(b)(6)** **NOTED ON MOTION CALENDAR: December 6, 2019** **ORAL ARGUMENT REQUESTED**

I, William Anthony Mason, declare as follows:

1. I am a technology expert retained on behalf of Plaintiff Royal J. O'Brien in the above-entitled action. I make this declaration based on personal knowledge, am over the age of 18 and competent to testify.

2. Attorneys for the Plaintiff have supplied me with a copy of the Complaint. I understand that the defendant, Microsoft, has moved for dismissal of that Complaint by urging that the Complaint does not meet the standard for pleading set forth in *Ashcroft v. Iqbal*, 556

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601 Union Street, Suite 4100
Seattle, Washington 98101-2380
(206) 628-6600

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U.S. 662 (2009), which was a United States Supreme Court case. As I understand that standard, the test is whether the Complaint states a plausible claim for relief. Determining whether a complaint states a plausible claim for relief will be a context-specific task that requires the reviewing court to draw on its judicial experience and common sense. In keeping with these principles, a court considering a motion to dismiss can choose to begin by identifying pleadings that, because they are no more than conclusions, are not entitled to the assumption of truth. While legal conclusions can provide the framework of a complaint, they must be supported by factual allegations. When there are well-pleaded factual allegations, a court should assume their veracity and then determine whether they plausibly give rise to an entitlement to relief.

3. So, a well-informed Court would review the Complaint to see if there are sufficient factual allegations, as opposed to mere conclusory assertions, to determine whether there has been a sufficient assertion of infringement. In the context of patent infringement, the Court will be guided by the Federal Circuit's decision in *Disc Disease Solutions Inc. v. VGH Solutions, Inc*., 888 F.3d 1256, 1260, (Fed. Cir. 2018). As I understand the case, the standard is one of whether the accused device is identified, a patent is asserted, and the defendant is the actor infringing under Section 271 of the Patent Act: "—Under *Iqbal/Twombly*, Disc Disease was required to 'state a claim to relief that is plausible on its face.' This plausibility standard is met when 'the plaintiff pleads factual content that allows the court to draw the reasonable inference that the defendant is liable for the misconduct alleged.' 'Specific facts are not necessary; the statement need only 'give the defendant fair notice of what the … claim is and the ground upon which it rests.''

4. I have been asked to review the Complaint to determine whether there are sufficient facts to give the defendant, Microsoft, fair notice of what the claim of infringement is

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601 Union Street, Suite 4100
Seattle, Washington 98101-2380
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and the ground upon which it rests. Because Microsoft can be assumed to have expertise in reading and evaluating the meanings of words that are unique to software and operating system operation, I have been asked to apply my expertise to evaluating the language of the Complaint. I have attached as Exhibit A to this Declaration, my Curriculum Vitae to demonstrate my expertise in the relevant field.

5. I state the following opinions to a reasonable degree of certainty:

a. The description of "cldflt.sys" in the original Complaint is one that would ordinarily be sufficient for someone of ordinary skill in the art of Windows kernel mode software development to identify the component being described.

b. That multiple clearly identifiable components in the current version of Windows 10 utilize the "Cloud Filtering API" (the term API means "application programming interface"), which is the mechanism that Microsoft has documented for using the functionality of "cldflt.sys".

6. To reach these opinions I took the following steps to determine the purpose of the "cldflt.sys" component. In doing this I verified its presence on my own system, identified documentation that Microsoft has made public about their Cloud Filtering technology, and verified the Microsoft components on my system that were utilizing it.

7. On Windows, the suffix ".sys" is normally associated with kernel mode drivers—which are dynamically loaded by the Windows kernel. These kernel mode drivers extend the functionality of the Windows operating system. Microsoft documentation explains that a .sys file is a special kind of dynamic link library: "The driver is the part of the package that provides the I/O interface for a device. Typically, a driver is a dynamic-link library (DLL) with the .sys file name extension. Long file names are allowed, except for boot-start drivers.

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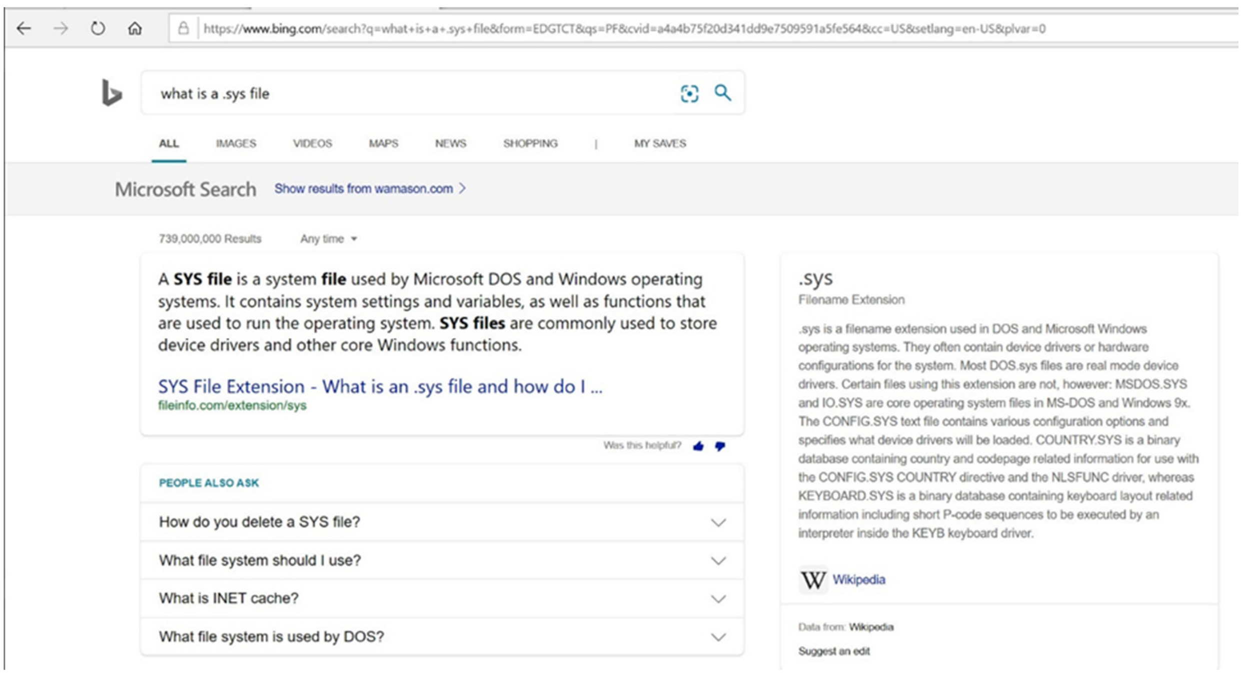
When a device is installed, Windows copies the .sys file to the \%SystemRoot%\system32\drivers directory" (Source: Microsoft, Components of a driver package, https://docs.microsoft.com/en-us/windowshardware/drivers/install/components-of-a-driver-package, [accessed November 10, 2019], May 2018.)

8. If I were unaware of what a ".sys" file does, I would simply type in a query to an Internet search engine of my choice; this is consistent with what one of ordinary skill in the art would use. For example, when I used the query "what is a .sys file" Microsoft's Bing search engine returned:



9. If I were unfamiliar with the functionality of "cldflt.sys", I would use a search engine to search for the specific name. Searching for a specific literal name is done by using quotation marks around the name in most search engines. For example, when I did this using Microsoft's Bing search engine, I received the following results:

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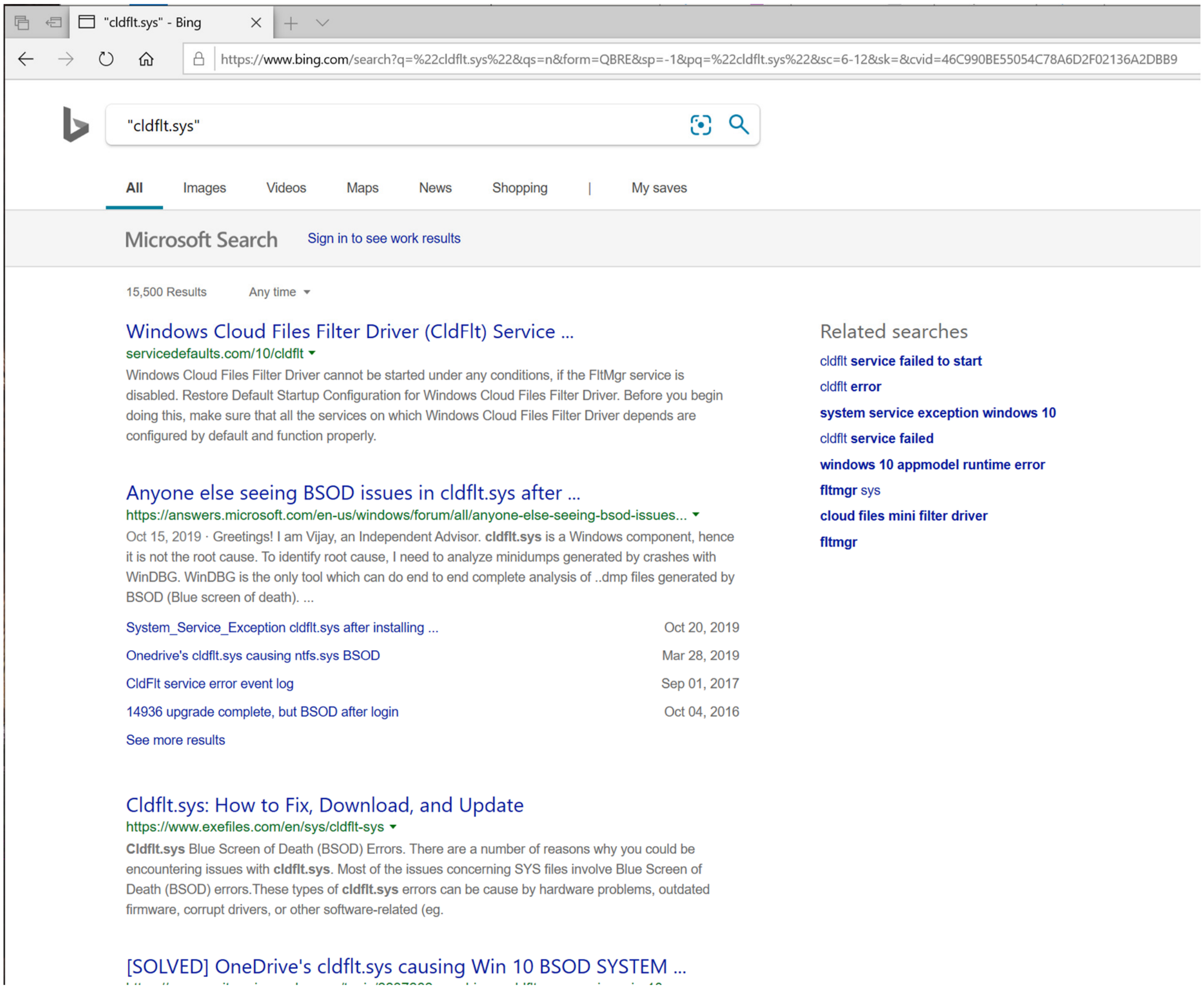

This would quickly alert me to the fact that this is a standard Microsoft-provided kernel device driver. Further, multiple links on the first search page returned identify that "cldflt.sys" is a Windows kernel driver that is associated with Microsoft's OneDrive service offering.

10. A search of Microsoft's public facing website quickly identifies the presence of the name cldflt.sys (https://docs.microsoft.com/en-us/windows-hardware/drivers/ifs/allocated-altitudes). From this, it becomes clear that this is a file system filter driver, registered to Microsoft Corporation.

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11. As a professional working in the field, I am familiar with this driver; it is a file system mini-filter driver that Microsoft has added to Windows to enable sparse storage of files. Specifically, when associated with a "cloud storage service" such as Microsoft's OneDrive product offering, it permits storage of a "stub" file that contains sufficient information for the contents of the file to be retrieved from a remote storage location as needed.

12. Microsoft incorporated support for "Cloud Sync Engines" beginning the 1709 release of Windows 10: "Starting in Windows 10, version 1709, Windows provides the cloud files API. This API consists of several native Win32 and WinRT APIs that formalize support for cloud sync engines and handles tasks such as creating and managing placeholder files and directories. Users of this API are typically sync providers and to some extent, Windows applications. (Source: Microsoft, Cloud filter reference, https://docs.microsoft.com/en-us/windows/win32/cfapi/cloud-filter-reference, [accessed November 10, 2019], May 2018.)

13. Most of the experts related to Microsoft file systems and Microsoft file system filter drivers are employees of Microsoft; it is reasonable to expect that were one to contact any of them, they would be able to quickly identify the specific technology component and even the development team responsible for it. Similarly, given the number of issues that I found while searching that led to system crashes ("Blue Screen of Death"), I would expect that the Microsoft product support team would be able to quickly identify the specific component. Thus, I was surprised that Microsoft's position is they were unable to identify the product based upon the description as one of ordinary skill in the art would have been able to quickly identify that this was a part of Windows.

14. I verified the presence of "cldflt.sys" on Windows by examining two separate systems: one is my Windows based workstation, which has Windows 10 Enterprise installed,

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and the second is a clean installation of Windows 10 Home from the October 2019 distribution ISO image downloaded from Microsoft's website.

15. I used the fltmc utility to list the filters. This is a standard administrative tool, distributed with Windows, that reports information about installed file system mini-filters (Source: Microsoft, Development and testing tools, https://docs.microsoft.com/en-us/windowshardware/drivers/ifs/development-and-testing-tools, [accessed November 11, 2019], Apr. 2017). This tool would be well-known to one of ordinary skill in the art of administering Windows computers.

16. The relevant output of this utility when I ran it on my own Windows-based computer system (Windows 10 1903 Enterprise):

```
PS C:\Windows\system32> fltmc filters

Filter Name                     Num Instances    Altitude    Frame
------------------------------  -------------  ------------  -----
bindflt                                 1       409800         0
FsDepends                               8       407000         0
WdFilter                                8       328010         0
storqosflt                              1       244000         0
wcifs                                   2       189900         0
PrjFlt                                  0       189800         0
dbx                                     2       186500         0
CldFlt                                  3       180451         0
FileCrypt                               0       141100         0
luafv                                   1       135000         0
npsvctrig                               1        46000         0
Wof                                     5        40700         0
FileInfo                                8        40500         0
```

I have highlighted the relevant line, showing the presence of "CldFlt". The driver corresponding to this is "cldflt.sys", a file system mini-filter driver.

17. I verified that this is a standard Windows feature and not something that might be present on my specific system by using a Windows 10 Home installation, installed into a virtual machine:

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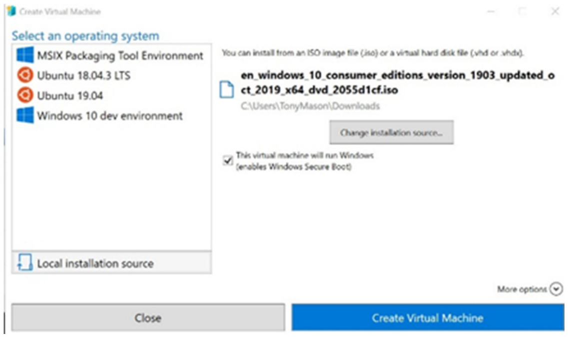
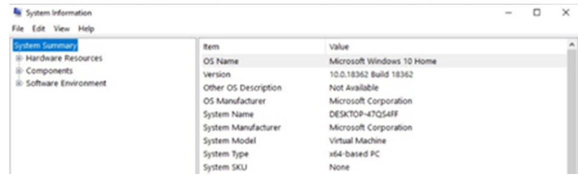
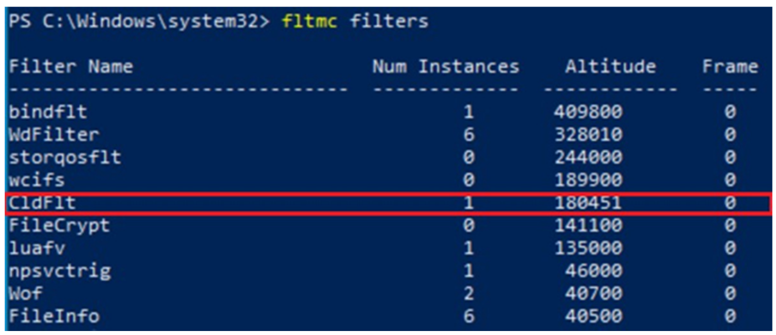


Hyper-V Installation



Installed Version Information



```
PS C:\Windows\system32> fltmc filters

Filter Name                     Num Instances    Altitude    Frame
------------------------------  -------------  ------------  -----
bindflt                                 1       409800         0
WdFilter                                6       328010         0
storqosflt                              0       244000         0
wcifs                                   0       189900         0
CldFlt                                  1       180451         0
FileCrypt                               0       141100         0
luafv                                   1       135000         0
npsvctrig                               1        46000         0
Wof                                     2        40700         0
FileInfo                                6        40500         0
```

fltmc filters information

Windows 10 1903 (October 2019 Update) distribution

Thus, for one of ordinary skill in the art of administering Windows systems, it is well known how to list the set of file system filter drivers present on the system.

18. Finally, it is well known on how drivers are loaded in Windows. For example, my book (P. Viscarola and W. A. Mason, Windows NT device driver development, New Riders Publishing, 1998) describes how the Windows registry is used to load such drivers. This basic mechanism has not fundamentally changed in the more than 26 years since Windows NT was released. Microsoft documents the details of how minifilters are installed into the registry as well, with the current documentation being available online (Microsoft,

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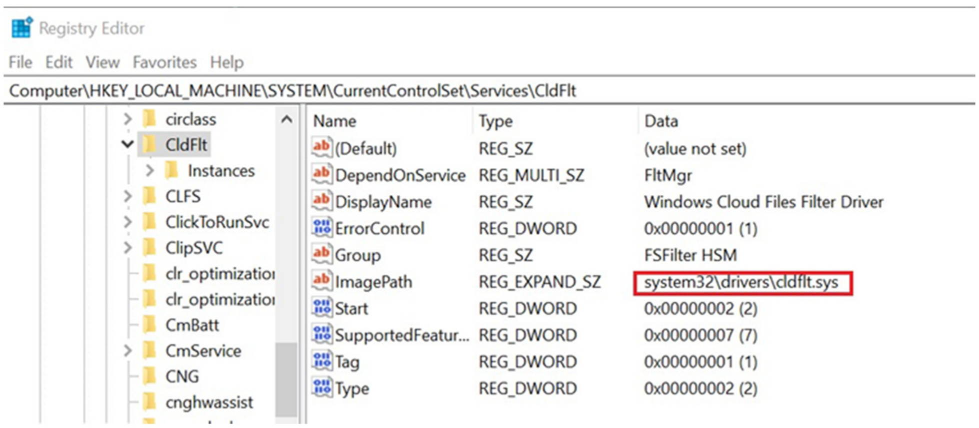
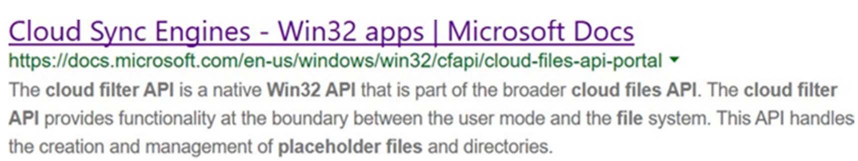
Installing a minifilter driver, https://docs.microsoft.com/en-us/windowshardware/drivers/ifs/installing-a-minifilter-driver, [accessed November 11, 2019], Apr. 2017).

19. I examined the Windows 10 registry to confirm that "CldFlt" was associated with "cldflt.sys":



Thus, for one of ordinary skill in the art of administering Windows systems, it is straight-forward to find "cldflt.sys" is part of a Windows 10 installation and that it is activated by default in Windows 10.

20. I note that the "DisplayName" listed in the Windows 10 registry associated with "CldFlt" is "Windows Cloud Files Filter Driver". I used this term in the Microsoft Bing Internet search engine. The first page of the returned results reference the "cloud filter API" on the Microsoft website:



When I followed this link, I found that it leads to extensive documentation on how to utilize the Cloud Filtering services (Source: Microsoft, Cloud sync engines,

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https://docs.microsoft.com/en-us/windows/win32/cfapi/cloud-files-api-portal, [accessed November 11, 2019], Feb. 2019).

21. The Microsoft Cloud Files API contains an extensive list of details about their library for exploiting this functionality. Further, I found the API call descriptions included specific details about the availability of these services. It is a standard pattern for Microsoft's technical documentation for Windows to list the specific libraries and Windows versions required to use these APIs:

Requirements

Minimum supported client	Windows 10, version 1709 [desktop apps only]
Minimum supported server	Windows Server 2016 [desktop apps only]
Target Platform	Windows
Header	cfapi.h
Library	CldApi.lib
DLL	CldApi.dll

(Source: https://docs.microsoft.com/en-us/windows/win32/api/cfapi/nf-cfapi-cfclosehandle)

Thus, I can see that this feature was first supported in Windows 10, Version 1709, as well as Windows Server 2016. I also note that this requires "CldApi.dll" be installed on the local system. This is a dynamic link library. I will use this information to identify services on Windows 10 that are exploiting this functionality.

22. I also noted that this documentation indicates the connection between "CldApi.dll" and "cldflt.sys": "The cloud filter API is a native Win32 API that is part of the broader cloud files API. The cloud filter API provides functionality at the boundary between the user mode and the file system. This API handles the creation and management of placeholder files and directories." (Source: Microsoft, Cloud sync engines,

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https://docs.microsoft.com/en-us/windows/win32/cfapi/cloud-files-api-portal, [accessed November 11, 2019], Feb. 2019). I also note that this documentation indicates that support for "sync engines" was present prior to Windows 10 1709, which is consistent with the text in the Complaint: "A sync engine" is a service that syncs files, typically between a remote host and a local client. Sync engines on Windows often present those files to the user through the Windows file system and File Explorer. Before Windows 10, version 1709, sync engine support in Windows was limited to scenario-agnostic ad-hoc surfaces, such as File Explorer's navigation pane, the Windows system tray, and (for more technical applications) file system filter drivers. Windows 10 version 1709 (also called the Fall Creators Update) introduced the cloud files API. This API is a new platform that formalizes support for sync engines. The cloud files API provides support for sync engines in a way that offers many new benefits to developers and end users. (Source: Microsoft, Build a cloud sync engine that supports placeholder files, https://docs.microsoft.com/en-us/windows/win32/cfapi/build-a-cloud-file-sync-engine, (accessed November 11, 2019), Feb. 2019).

23. Thus, I conclude that this new Cloud Filtering API was introduced to make use of the cloud filtering services exported by the "cldflt.sys" file system mini-filter driver easier to implement and use by application programs. I also conclude that an application would generally not use this unless it were using one or more of the services provided by the "cldflt.sys" file system mini-filter driver.

24. I identified that "CldApi.dll" would be used by applications that are in turn using the services of the "Windows Cloud Files Filter Driver" (also known as "cldflt.sys"). I used the Microsoft provided utility Process Explorer (available from https://docs.microsoft.com/en-us/sysinternals/downloads/process-explorer) to examine the

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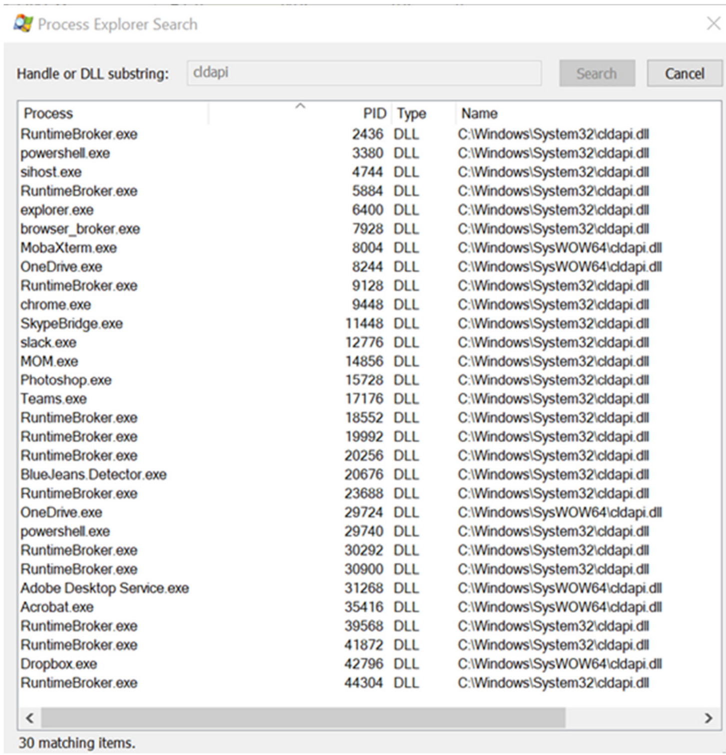
software components running on my own Windows 10 1903 system to determine which of them were using the "CldApi.dll" dynamic link library:



Process Explorer Search

Handle or DLL substring: cldapi

Search Cancel

Process	PID	Type	Name
RuntimeBroker.exe	2436	DLL	C:\Windows\System32\cldapi.dll
powershell.exe	3380	DLL	C:\Windows\System32\cldapi.dll
sihost.exe	4744	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	5884	DLL	C:\Windows\System32\cldapi.dll
explorer.exe	6400	DLL	C:\Windows\System32\cldapi.dll
browser_broker.exe	7928	DLL	C:\Windows\System32\cldapi.dll
MobaXterm.exe	8004	DLL	C:\Windows\SysWOW64\cldapi.dll
OneDrive.exe	8244	DLL	C:\Windows\SysWOW64\cldapi.dll
RuntimeBroker.exe	9128	DLL	C:\Windows\System32\cldapi.dll
chrome.exe	9448	DLL	C:\Windows\System32\cldapi.dll
SkypeBridge.exe	11448	DLL	C:\Windows\System32\cldapi.dll
slack.exe	12776	DLL	C:\Windows\System32\cldapi.dll
MOM.exe	14856	DLL	C:\Windows\System32\cldapi.dll
Photoshop.exe	15728	DLL	C:\Windows\System32\cldapi.dll
Teams.exe	17176	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	18552	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	19992	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	20256	DLL	C:\Windows\System32\cldapi.dll
BlueJeans.Detector.exe	20676	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	23688	DLL	C:\Windows\System32\cldapi.dll
OneDrive.exe	29724	DLL	C:\Windows\SysWOW64\cldapi.dll
powershell.exe	29740	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	30292	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	30900	DLL	C:\Windows\System32\cldapi.dll
Adobe Desktop Service.exe	31268	DLL	C:\Windows\SysWOW64\cldapi.dll
Acrobat.exe	35416	DLL	C:\Windows\SysWOW64\cldapi.dll
RuntimeBroker.exe	39568	DLL	C:\Windows\System32\cldapi.dll
RuntimeBroker.exe	41872	DLL	C:\Windows\System32\cldapi.dll
Dropbox.exe	42796	DLL	C:\Windows\SysWOW64\cldapi.dll
RuntimeBroker.exe	44304	DLL	C:\Windows\System32\cldapi.dll

30 matching items.

Process Explorer identifies programs using the Cloud Filter API dynamic link library. Using this list, I identified the components using the Cloud Filter API", which enables the use of the "cldflt.sys" file system mini-filter driver that is named in the complaint, and are associated with Microsoft-provided software:

RuntimeBroker.exe	Windows 10	Windows service provider
powershell.exe	Windows 10	Command line interpreter/shell
sihost.exe	Windows 10	Shell Infrastructure Host (part of graphical interface)
explorer.exe	Windows 10	Windows 10 Graphical File Explorer

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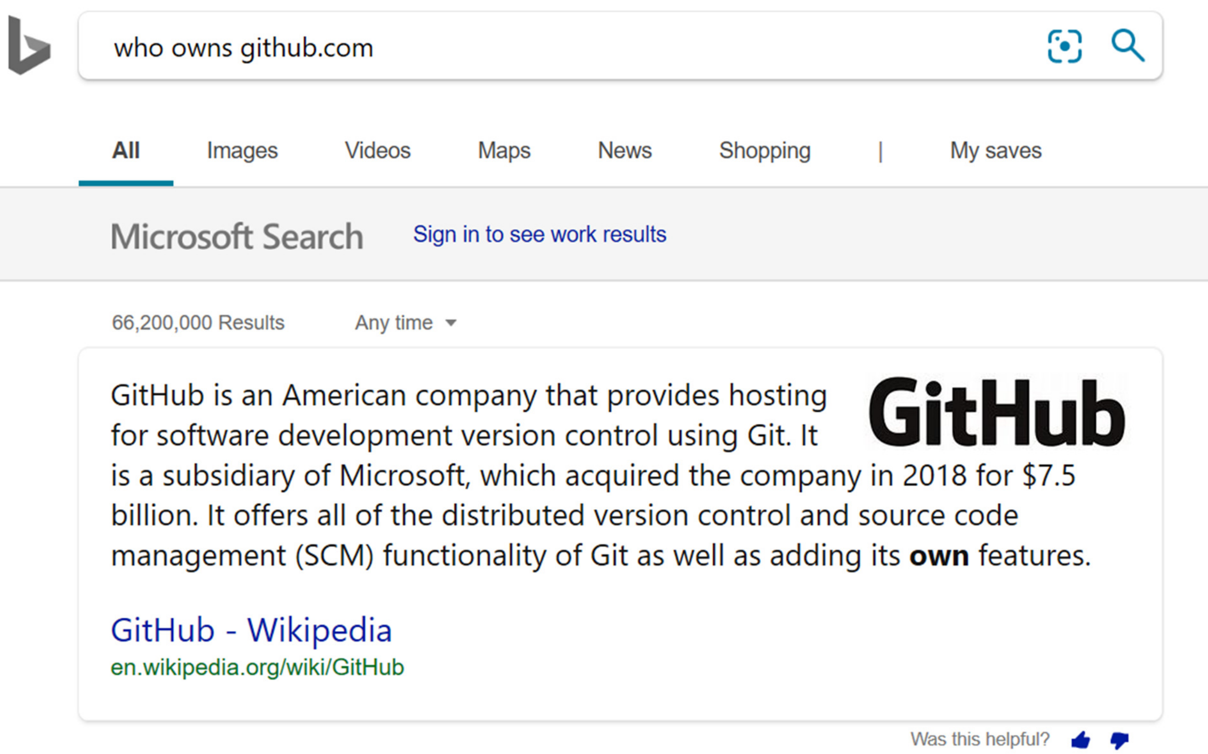
OneDrive.exe	Windows 10/Office 365	Cloud Storage Provider
SkypeBridge.exe	Skype	Microsoft telecommunications product
Teams.exe	Office 365	Business team communications software

Microsoft provides ample documentation of the functionality of the Cloud Filter API. Further, Microsoft provides sample code for its use (Microsoft, Cloud mirror sample, https://github.com/Microsoft/Windows-classicsamples/tree/master/Samples/CloudMirror, [accessed November 11, 2019], Mar. 2019). I note that Github.com, the site hosting the code samples, belongs to Microsoft.



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In reviewing the code present in that repository (where a "repository" is a unit of information management on Github and represents a collection of files), I can see that there are multiple people that appear to be Microsoft employees that have, within the past month, contributed code changes to this sample code. I was not able to ascertain, based upon the documents provided to me, if Microsoft's counsel had consulted with these experience Microsoft developers.

25. The original Complaint clearly identifies both "cldflt.sys" and "cldapi.dll". It is easily determined from the available public Microsoft documentation the specific functionality that these components provide. When I perused the public facing documentation it provided extensive descriptions of the functionality provided by these Windows 10 components. While Windows 10 does provide extensive functionality, it does so in a highly modular format. The original complaint, by clearly calling out the Cloud Filtering functionality identified the distinctly identifiable components of Windows 10 that are alleged in the C omplaint. Thus, the identification is specific, clear, and targeted towards one important and clearly identifiable functional component of Windows.

26. Thus, my conclusion: there is no question that the minifilter "cldflt.sys" has been accused. I read the motion and understand that Microsoft does admit that "cldflt.sys" is suitably identified as an element of Windows 10 so I started from there. Microsoft, being familiar with their own product at a deep technical level, would be able to clearly follow a trail of logic similar to the one that I followed. They would know that "cldflt.sys" contains the algorithmic logic and configuration information that enables Windows to provide specific functionality. Since the Complaint sets forth the contention that "cldflt.sys" is practicing the patent, it seems unambiguous to me; I would expect it to be similarly unambiguous to one of ordinary skill in the art.

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THE FOREGOING IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE, SO STATED UNDER PENALTY OF PERJURY FOR THE STATE OF WASHINGTON.

DATED this 21st day of November, 2019 at Vancouver, British Columbia.

William Anthony Mason

Williams, Kastner & Gibbs PLLC
601 Union Street, Suite 4100
Seattle, Washington 98101-2380
(206) 628-6600

CERTIFICATE OF SERVICE

I hereby certify under penalty of perjury under the laws of the State of Washington that I electronically filed the foregoing with the Clerk of the Court using the CM/ECF system, which will send electronic notification of such filing to all CM/ECF participants.

DATED this 2nd day of December, 2019.

s/ Mark Lawrence Lorbiecki
Mark Lawrence Lorbiecki, WSBA #16796

Williams, Kastner & Gibbs PLLC
601 Union Street, Suite 4100
Seattle, Washington 98101-2380
(206) 628-6600

6996730.1

EXHIBIT A

Tony Mason
Curriculum Vitae

+1 (888) 678-9891
tony@wamason.com
https://wamason.com
https://www.linkedin.com/in/tonymason2

DOCTORAL RESEARCH

"Percipience: Enabling Personal Data Mining"

My research examines mechanisms for extending traditional file systems functionality to enable the capture, storage, retrieval, and search of the relationships between data elements. Because of the vast body of existing applications, my work explores how to support existing applications at the same time enabling incremental evolution of applications to exploit this functionality to provide a richer set of tools to find data relationships amongst the user's personal data.

WORK EXPERIENCE

CURRENT, FROM JULY 2017 (PT)

wamason.com
Expert

Litigation support for clients requiring expert services. This work primarily focuses on assisting in the preparation of materials necessary for patent litigation. I provide research, prepare reports, explain technical terms clearly and lucidly, engage in depositions, etc. The focus is on clearly explaining potentially complex technical concept and terms to non-technical people.

CURRENT, FROM AUGUST 2015 (PT)

Georgia Institute of Technology
Instructional Assistant

Provide support for the Online Master of Science in Computer Science programs course. Responsibilities included providing formative and summative analysis to students, develop automatic grading tools, explain concepts to students, provide direction to other team members, work with instructors regarding course execution and improvements. Responsible for detecting and reporting student plagiarism.

CURRENT, FROM 2011 (PT)

BitRaider
Technical Adviser

Provide technical advice to BitRaider, a company specializing in streaming game delivery.

NOVEMBER 1994 – NOVEMBER 2016 (FT)

OSR Open Systems Resources, Inc.
Consulting Partner and Treasurer/Vice-President

In this role I was responsible for providing services to commercial customers, by designing and implementing key file systems technologies for file systems and file system filter drivers. I was involved in critical aspects of our educational offerings, teaching classes in Driver Development, File Systems Development, File System mini-filter development and Kernel Debugging. Focus was on the Windows OS platform. Recent projects include: Isolation Filter Kit that provides multiple simultaneous views of

EDUCATION

2017 – PRESENT	**Doctor of Philosophy** IN PROGRESS Computer Science *University of British Columbia*
2015 – 2017	**Master of Science** Computer Science *Georgia Institute of Technology*
1983 – 1987	**Bachelor of Science** Department of Mathematics *University of Chicago*

COMPUTER SKILLS

LANGUAGES	C, C++, Python, LaTeX, etc.
OPERATING SYSTEMS	Windows, Linux, UNIX, etc.

COMMUNICATION SKILLS

CONFERENCES	Invited Speaker at WDC Invited Speaker at IDC Invited Speaker at AFSUG Retrospective
POSTERS	Percipience - UBC CS-50 – 2018
COURSES	Developing Windows File Systems Developing Windows FS Mini-Filters Advanced Windows Driver Development Windows Internals (Source Code) Porting Windows Drivers to 64-bit Windows Kernel Debugging

SKILLS

Goal Oriented

I believe in action over long-winded discussions. I listen to everyone's viewpoints and use my judgement to act based on consensus to achieve goals quickly and efficiently.

Fail Fast

When working on projects it is important to work towards success. One way I use to achieve this is to establish mechanisms to tell *when* something is not going to work — to *fail fast* in order to maximize the useful work and minimize the cost of investigating paths that will not lead to success.

Education

One of the most rewarding aspects of my work is in helping other people learn and grow. It is deeply rewarding when former students, employees, and colleagues tell me how I helped a single file on either local or network storage without modifying the underlying file system. This abstraction was useful for transparent per-file encryption in that different applications could be given different views (raw or encrypted) of the file, with coherency, specialized file system mini-filter for transparently backing up SQL databases to Azure, another for doing block level change tracking for SQL databases.

SEPTEMBER 1993 TO NOVEMBER 1994 (FT)

FORE Systems
Technical Manager

In this role I was responsible commercial ATM adapter device driver development for PC platforms (Windows, Novell, and OS/2). Actively participated in the ATM Forum (SA&A and LAN Emulation SWG). Worked with both Microsoft and Novell to define and establish architectural support for ATM.

JULY 1989 TO SEPTEMBER 1993 (FT)

Transarc Corporation
Area Manager (previously Member of Technical Staff)

Technical management for DCE LFS physical media file systems work: coordinate activities of team members, managed DCE/DFS integration including working with key external partners: OSF, HP, IBM and Digital Equipment Corporation. Transarc AFS development on various UNIX platforms, DCE/DFS development. As a member of the Episode team I was responsible for designing and implementing the transactional system, still novel for its use of undo/redo model journaling. Part of the DCE/DFS development team, participated in architecuture and design.

JUNE 1987 TO JUNE 1989 (FT)

Leland Stanford Jr. University
Research Staff

Working in Professor David Cheriton's Distributed Systems Group. Responsible for systems administration for the group. Implemented the UNIX version of VMTP (Versatile Message Transport Protocol), a reliable transport protocol for use over IP, including IP multicast. Wrote kernel mode device drivers for Ethernet NICs for the V Operating System, as well as a primitive kernel debugger.

them in improving their own work. To achieve this, I strive to break things down and make them understandable based upon the listener's current level of understanding.

Passionate

I have been involved in computer systems from an early age. My education, work, and research have fueled my passion for these topics. I enjoy learning new technologies and finding ways to re-examine old technologies in light of greater understanding. I greatly enjoy finding new ways to make systems better, not only in terms of performance but also as tools for creativity and productivity for *all* users.

PATENTS

US/9830329 METHODS AND SYSTEMS FOR DATA STORAGE
US/9600486 FILE SYSTEM DIRECTORY ATTRIBUTE CORRECTION
US/9535759 WORK QUEUE THREAD BALANCING
US/8990228 ARBITRARY DATA TRANSFORMATION
US/8903874 FILE SYSTEM DIRECTORY ATTRIBUTE CORRECTION
US/8539228 MANAGING ACCESS TO A RESOURCE
US/8521752 ARBITRARY DATA TRANSFORMATIONS
US/8024433 MANAGING APPLICATION RESOURCES
US/7809897 MANAGING LOCK RANKINGS
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US/7949693 LOG-STRUCTURED HOST DATA STORAGE
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The Episode File System, USENIX Technical Conference Winter 1992. (Co-author)

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